This piece is a science highlight written specifically for this application, based on a paper recently published at the ALS. I chose it because it is concise enough to summarize fully within a writing sample and because it gave me the opportunity to demonstrate my ability to communicate work directly relevant to the ALS mission. The piece is accompanied by five original illustrations I created to support the text.

Before submitting, I reached out to lead author Dr. Kat Armstrong and asked her to review for factual accuracy. She responded with one correction and one suggested improvement, both of which I incorporated: the heater wire material, which I had incorrectly identified as platinum-rhodium (the correct composition of the thermocouple, not the heater coil), and the pressure analogy in the opening paragraph, which I sharpened using her suggested comparison: the weight of the Eiffel Tower on a surface the size of a golf ball.

A Hot Take on a Hard Squeeze

A simple wire-coil heater opens a new temperature window for high-pressure X-ray experiments at the ALS

S.J. Kranz
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A highlight of "A simple wire-coil resistive heater for high temperature radial x-ray diffraction in a diamond anvil cell" by K. Armstrong, et al. *Rev. Sci. Instrum*. 97, 013905 (2026).

Diamond anvil cells are crushing it

Squeezing things is super useful: juicing a lemon, getting that last bit of toothpaste, or relieving stress with an inflatable ball. But sometimes you need something thousands of times stronger than your hand—like a pneumatic press, which stamps metal for car parts or pasteurizes your guacamole without heat. But what if you want to study materials under the pressures found deep inside the Earth? That's where diamond anvil cells (DACs) come in. When a sample is placed between the tips of two diamonds and squeezed, the force concentrates onto a microscopic area, generating enormous pressure upwards of hundreds of gigapascals. To put that in perspective, 25 GPa is roughly equivalent to balancing the weight of the Eiffel Tower on a surface the size of a golf ball. Shine the Advanced Light Source's intensely bright X-rays at that sample and you can measure its atomic structure and material properties with extraordinary precision.

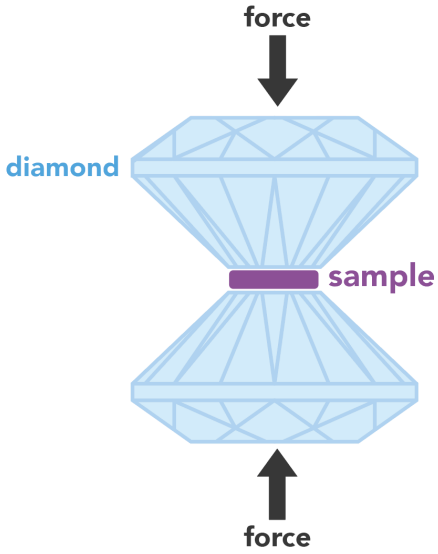


Fig 1. A simplified diagram of a diamond anvil cell (DAC). Typically the diameter of the tips of the diamonds are around 100 to 250 μm. Adapted from Wikipedia.

Two ways to look at a squeeze

Scientists use DACs in two different orientations depending on what they want to know: axial and radial. In axial geometry, the X-ray beam travels along the same axis as the squeeze, essentially looking straight down the barrel of the press. This tells you a material's crystal structure and how it compresses: which atomic arrangement it prefers, whether it undergoes a phase transition, and how its volume changes with pressure. It's the workhorse of high-pressure science.

Radial geometry asks a different question. Here the beam comes in from the side, perpendicular to the squeeze, watching the material get squished from the sidelines rather than from above. That sideways view captures something axial geometry fundamentally can't: the directionality of the stress, and how hard the material resists it. This makes radial DAC the tool of choice for measuring yield strength, hardness, and deformation behavior. And this insightful utility is why Dr. Kat Armstrong and others at the ALS decided to develop an improvement for this experimental technique.

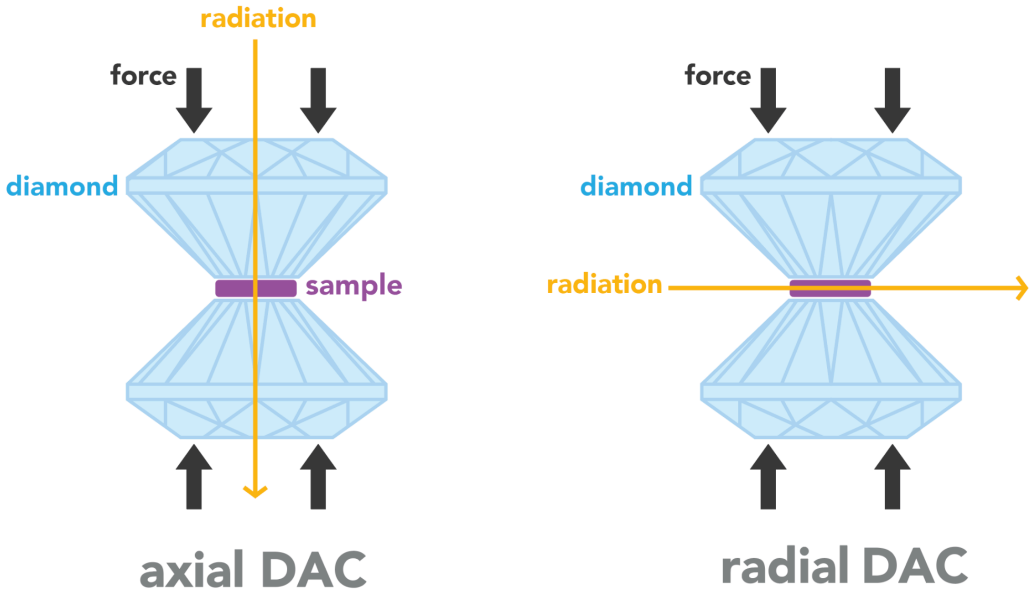


Fig 2. In the axial configuration of a DAC, the probing radiation travels parallel to the axis of the applied force. In radial DAC, the radiation is perpendicular to the force.

It's getting hot in here

The deep interiors of planets aren't just high-pressure environments. They're also hot. Like, really hot. So doing these compressive experiments at a high temperature is pretty handy. Adding controlled heat to an axial DAC is relatively straightforward: wrap a small wire coil around the diamond tips like a tiny electric burner, and you're in business. But in radial geometry, that coil sits directly in the path of the incoming X-ray beam and the outgoing diffracted signal, which blocks the very view you're trying to capture. Previous workarounds either used lasers, which heat unevenly (temperatures can vary by hundreds of degrees across fractions of a millimeter!) or on exotic graphite foil setups that are difficult to replicate. The result: high-temperature radial diffraction has remained stubbornly out of reach for most laboratories. But a team working at ALS beamline 12.2.2 has a solution.

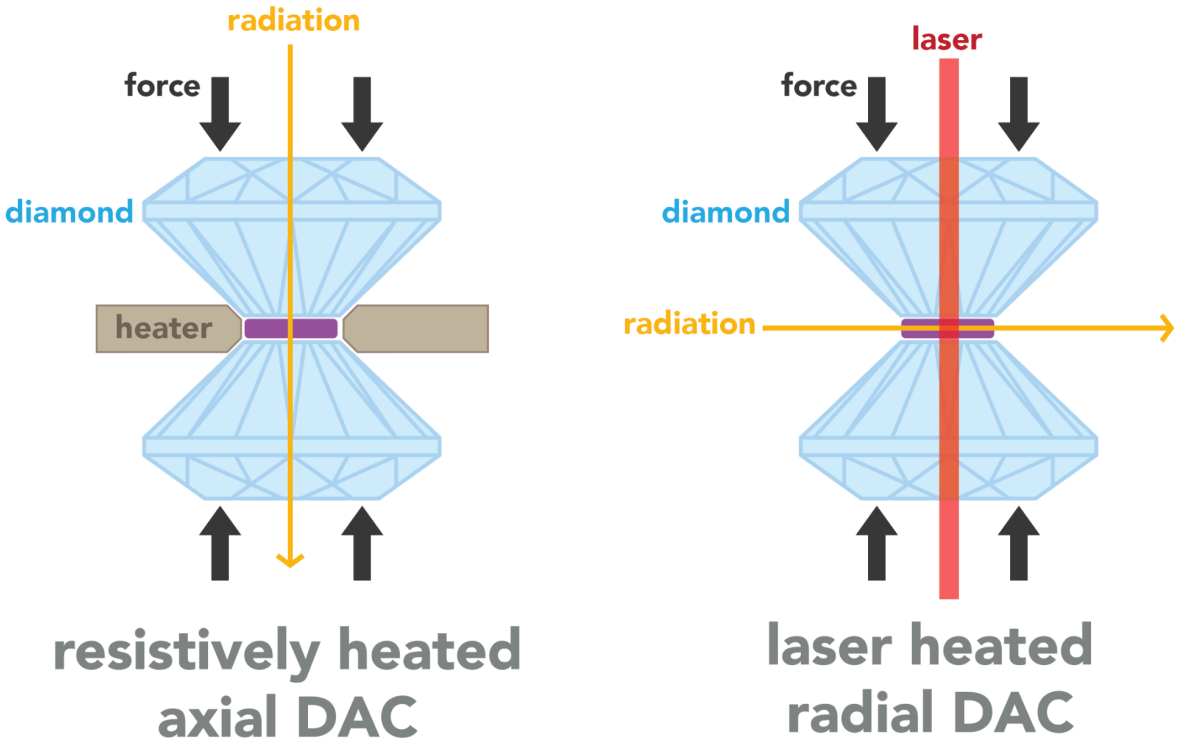


Fig 3. A ring-shaped resistive heating element can evenly heat a sample in an axial DAC setup without interfering with the probing radiation. However, in a radial DAC, the heating element would be in the way of the radiation. Laser heated setups avoid this interference, but they have drawbacks, such as large temperature variations across the sample.

A low-tech solution for a high-pressure problem

The solution is an elegant modification of existing infrastructure. Resistive wire-coil heaters for DACs are already sophisticated devices: precision-machined ceramic discs wound with platinum wire and insulated with mica. This heater design was developed by a group at the University of Hawai'i, described in detail by Lai et al. (2020) and Wang et al. (2023), and had already proven itself in axial geometry experiments. Armstrong et al. obtained the pyrophyllite discs directly from that lab and got to work. Their key insight was geometric: in two places, they rerouted the heater wires into arcs and cut corresponding windows into the ceramic disc, opening clear channels for X-rays to enter and diffracted signals to exit. The modification itself is straightforward. The insight was recognizing that so little modification was sufficient.

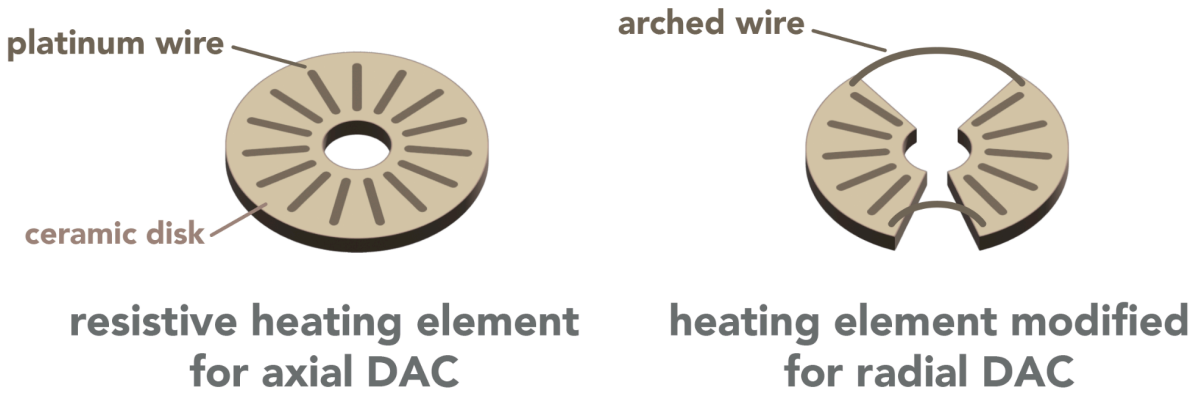


Fig 4. Left, a diagram of a disk-shaped resistive heating element designed for use in axial DAC (adapted from Lai, et al. 2020). Right, a simple modification of the heater removes sections to allow for x-rays to enter and exit the sample unimpeded. Arched wires bridge the removed sections of the disk.

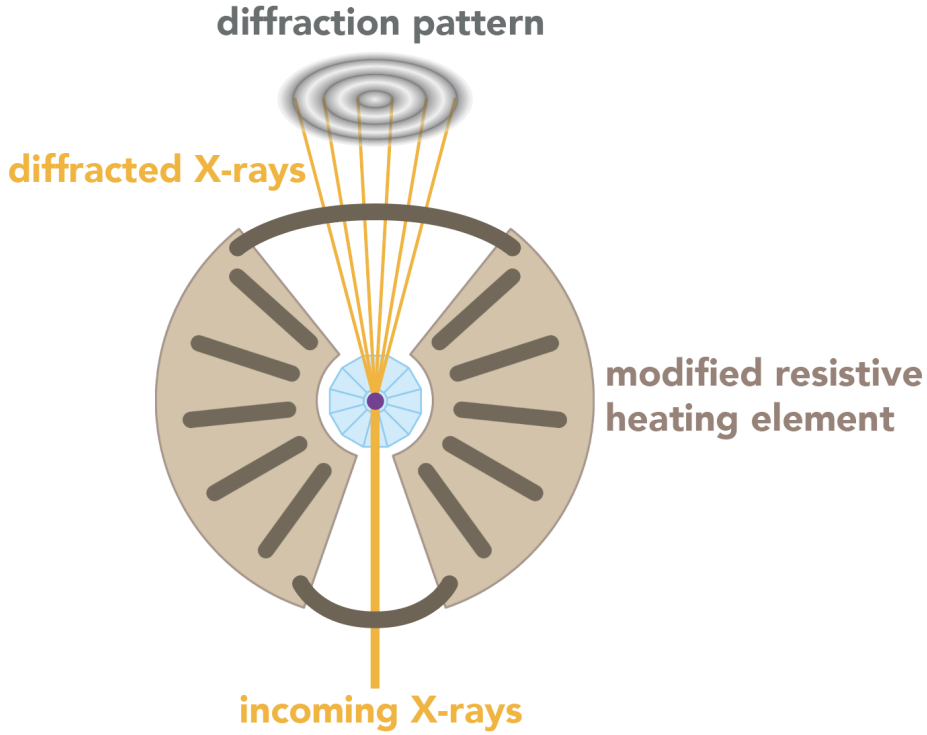


Fig 5. The resistive heating element is used for radial DAC. The diamond anvil (blue) and sample (purple) are surrounded by the heating element. The gaps in the heater allow do not impede X-rays and the exiting diffracted X-rays, thus allowing a diffraction pattern to be captured. In this diagram the axis of the force is into the page.

Location, location, location

That leads us to the age-old questions in scientific experimentation: Does it actually work? And how can we prove it? A heater with windows cut into it is only useful if it still heats the sample evenly, and if the thermometer is telling the truth. To check, the team ran calibration experiments with two thermocouples simultaneously: one where the sample would sit, and one mounted on the diamond surface as in a real experiment. The verdict: the two thermocouples agreed within 10 degrees when properly aligned. The operative word being "properly." If the heater shifts enough that the diamond tips fall outside its inner ring, the temperature readings diverge by 200 degrees. Good data, in this case, depends as much on careful setup as on clever design.

What did they find?

The novel setup delivered. Armstrong et al. squeezed tungsten carbide, a material thoroughly characterized at high pressure and thus ideal for a benchmark, to pressures up to 60 GPa while simultaneously heating it to 300°C and collecting radial diffraction data. The results confirmed what prior studies had found: tungsten carbide softens as it heats up, and the degree of softening depends on which direction through the crystal you're measuring. Crucially, those results matched previously published hardness data going back decades. When you're validating a new technique, reproducing the expected result isn't anticlimactic: it's the goal. A technique that reproduces known results under new conditions is a technique you can trust.

The broader implication is about access. High-temperature radial diffraction has long been a niche capability, limited by the geometric headaches described above. This adaptation, built on commercially available equipment, using a previously-designed heater, puts that capability within reach of the general user community at ALS. For researchers studying how materials behave under the conditions they actually encounter in nature, that's not a small thing. The moderate temperature range this technique unlocks—too hot for room-temperature measurements, too cool for laser heating—is precisely where geologically relevant processes live: mineral deformation, mantle flow, the warm-but-not-molten interiors of rocky planets. The window is now open.